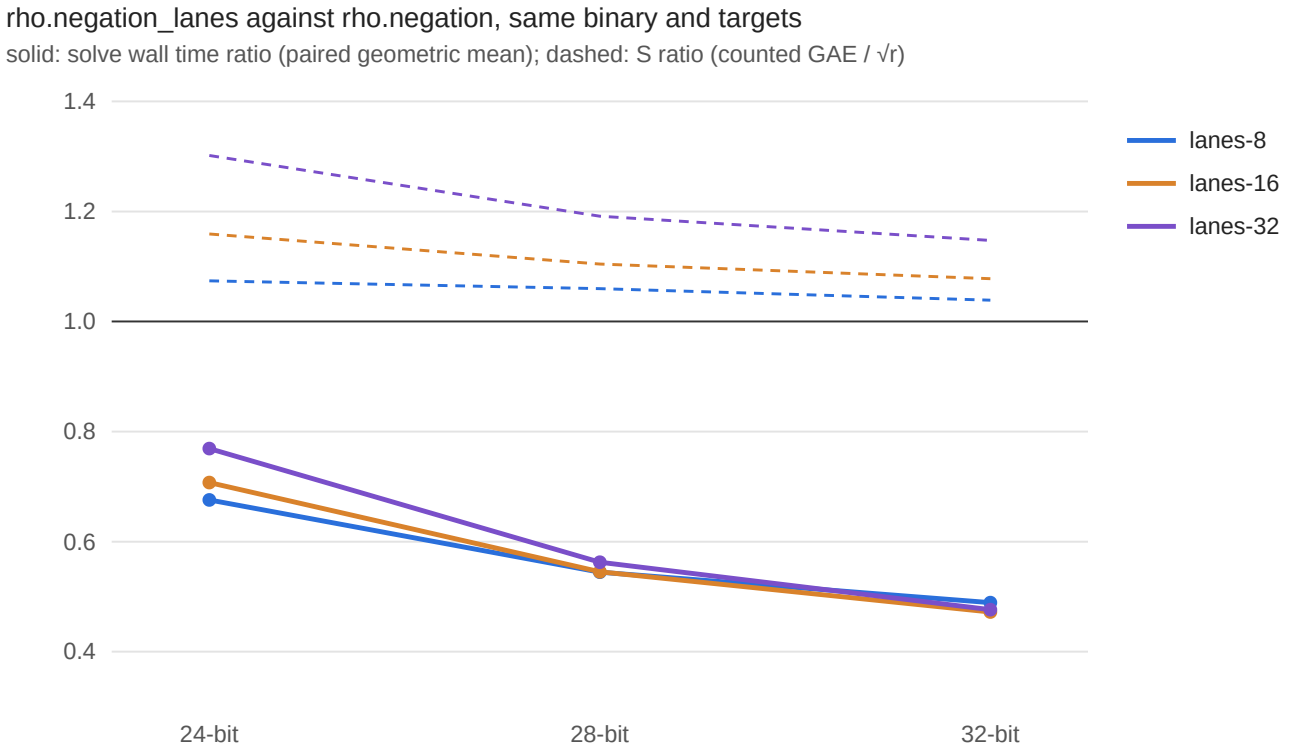
rho.negation_lanes: Pollard rho with batched inversions (2026-10-23)

A new ecbench method beside `rho.negation`, whose code and counts are unchanged. It runs `rho.negation`'s step rule (negation classes, look-ahead, cycle escape, distinguished points) as `lanes` walks in lockstep on a prime curve; each round's additions share one field inversion (Montgomery's trick). Every addition any lane performs is charged, including the walks still in flight when another lane's collision recovers the logarithm, so S is a few percent above `rho.negation`'s. The `field_ops` block records the work actually done: 3(n − 1) multiplications and one inversion per round of n additions, then 2M + 1S per addition, against rho.negation's 2M + 1S + 1I.



bits	arm	verified	mean S	median wall ms	wall / rho (paired)	S / rho (paired)	M / S / I per GAE
24	rho-neg	160/160	1.083	0.72	1.000	1.000	2.15 / 1.05 / 1.000
24	rho-neg-aa	160/160	1.083	0.73	1.001	1.000	2.15 / 1.05 / 1.000
24	lanes-8	160/160	1.158	0.50	0.676	1.074	4.52 / 1.05 / 0.209
24	lanes-16	160/160	1.228	0.53	0.707	1.159	4.68 / 1.04 / 0.149
24	lanes-32	160/160	1.349	0.59	0.769	1.302	4.77 / 1.04 / 0.117
28	rho-neg	160/160	1.002	2.67	1.000	1.000	2.04 / 1.01 / 1.000
28	rho-neg-aa	160/160	1.002	2.64	1.019	1.000	2.04 / 1.01 / 1.000
28	lanes-8	160/160	1.045	1.44	0.544	1.060	4.58 / 1.01 / 0.153
28	lanes-16	160/160	1.080	1.44	0.545	1.105	4.77 / 1.01 / 0.092
28	lanes-32	160/160	1.156	1.43	0.563	1.191	4.86 / 1.01 / 0.061
32	rho-neg	120/120	0.930	9.24	1.000	1.000	2.02 / 1.01 / 1.000
32	rho-neg-aa	120/120	0.930	9.23	0.987	1.000	2.02 / 1.01 / 1.000
32	lanes-8	120/120	0.959	4.57	0.489	1.039	4.61 / 1.01 / 0.135
32	lanes-16	120/120	0.989	4.44	0.472	1.078	4.79 / 1.01 / 0.074
32	lanes-32	120/120	1.045	4.45	0.476	1.147	4.89 / 1.00 / 0.043

Four sessions per curve on one binary, every arm alternated within each session over the same targets and rounds; warm-up excluded. The A/A control (`rho-neg-aa`) is a second copy of the reference arm. Every session replays exactly (`ecbench verify --replay-all`, 2200 replays). At 32 bits a rho step's single field inversion (word extended Euclid, about 110 ns) was most of its wall time; shared over a round it costs a few nanoseconds, and the step becomes instruction-bound (about 680 instructions per group operation in Callgrind, for both methods).

Prime index calculus: prefetch for lane walks

The prime lane target sources prepare a round of query points before taking the first; the collector now reads the guard set's and the m = 2 pair table's slots for the whole round first, uncounted, so their cache misses overlap. Old and new binary, three sessions each on the 32-bit curve (`targets=walk-prime-batch-16`); 108 of 108 records identical outside timing, every session replays under the other binary.

arm	ns/trial old	new	ratio	solve ms old	new	ratio
b2048-blocked-b16	464	377	0.812	413	401	0.971
b512-batch16	330	309	0.935	632	588	0.930

This is a wall-time result. In the counted unit (S, GAE/√r) the lanes method costs 4–30% more than `rho.negation`, so it is not a stronger matched rho reference for index-calculus comparisons, which are made in that unit.